

Geospatial Segmentation via Embedding-driven Active learning for Mask generation

Author: Igor Vons (igorvons3@gmail.com) | Supervisors: Mikel Galar (mikel.galar@unavarra.es) & Christian Ayala (cayala@itracasa.es)

Context & Motivation

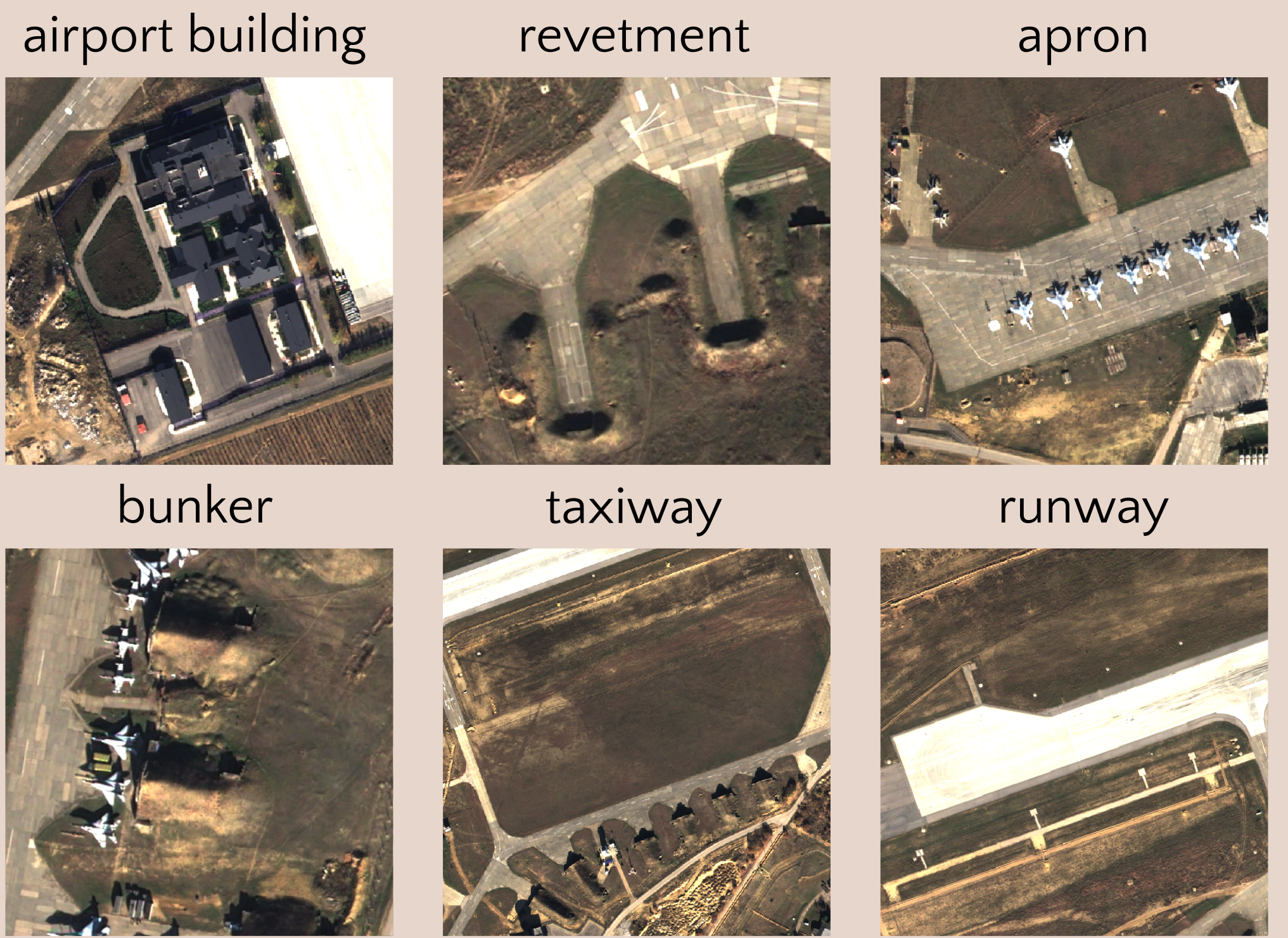
High-resolution satellite imagery is widely available, yet deploying dense semantic segmentation models in specialized or security-sensitive domains remains bottlenecked by the complete **absence of curated, pixel-level training datasets**. While **crowdsourced spatial data** like OpenStreetMap (OSM) offers global coverage, it is **often outdated, incomplete, and poorly aligned**. As a result, directly training deep learning architectures on these weak labels propagates confirmation bias, **forcing downstream models to memorize historical errors rather than actual visible structures**.

Objective

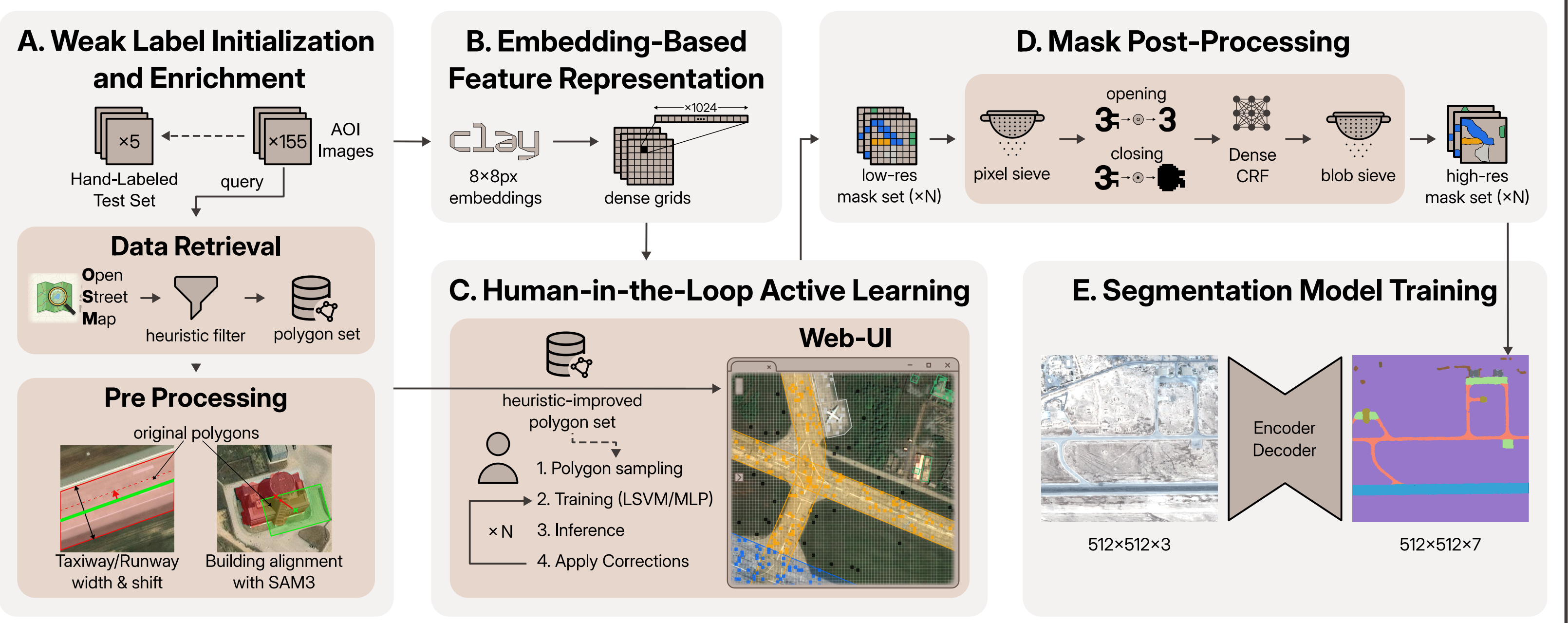
Build a **scalable framework** that transforms **raw satellite imagery and noisy annotations into reliable, high-quality segmentation masks** that can be used to train task-specific models with **minimal manual supervision**.

Dataset

Comprises **155 Very High Resolution** (0.3-0.75m GSD) Pléiades **satellite images** acquired across **globally distributed airbases**. Depending on the specific sensor, imagery includes **Near-Infrared** band, safe for some scenes that only feature standard **RGB** bands. The dataset is designed to evaluate classification performance across **7 distinct airbase land-use categories**.



Proposal: Geo-SEAM



Geo-SEAM (Geospatial Segmentation via Embedding-driven Active-learning for Mask generation) is a **modular human-in-the-loop framework that turns raw imagery and noisy vectors into high-fidelity datasets** through five core stages:

A. Weak Label Initialization and Enrichment: Obtain and fix messy OSM geometries using spectral indices (NDVI/VARI) and SAM zero-shot prompts.

B. Embedding-Based Feature Representation: Map imagery into a robust feature space using the Clay Geospatial Foundation Model.

C. Human-in-the-Loop Active Learning: Use a fast lightweight classifier on the resulting embedding grid to fill and improve the class

D. Mask Post-Processing: Smooth down patch predictions using DenseCRF to create clean pseudo-ground truth for training.

E. Segmentation Model Training: Use the obtained masks to train and evaluate a semantic segmentation model.

Integrated Active Learning Web UI Features

Centralized workspace | Easy image browsing | Polygon & freehand sampling
Live visual corrections | Prediction & map overlays | Batch training queues

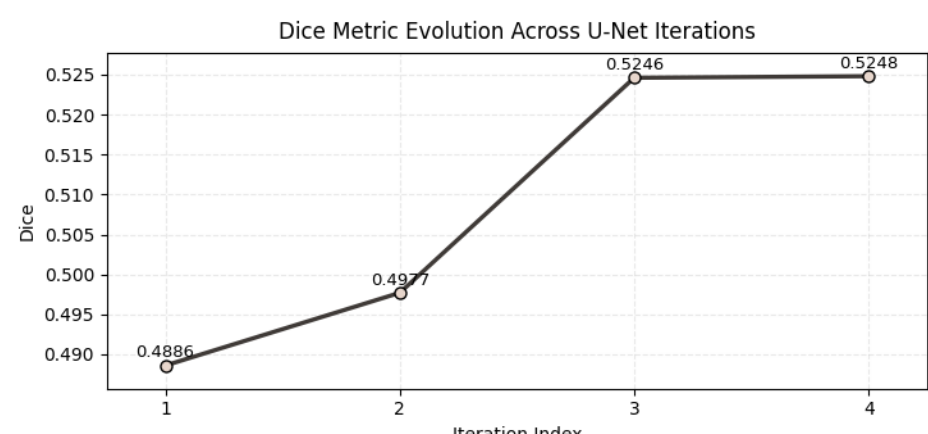


Results & Evaluation

RQ1. How many active learning iterations are needed?

U-NET ITERATION BENCHMARK FOR ResNet-34 ARCHITECTURES.

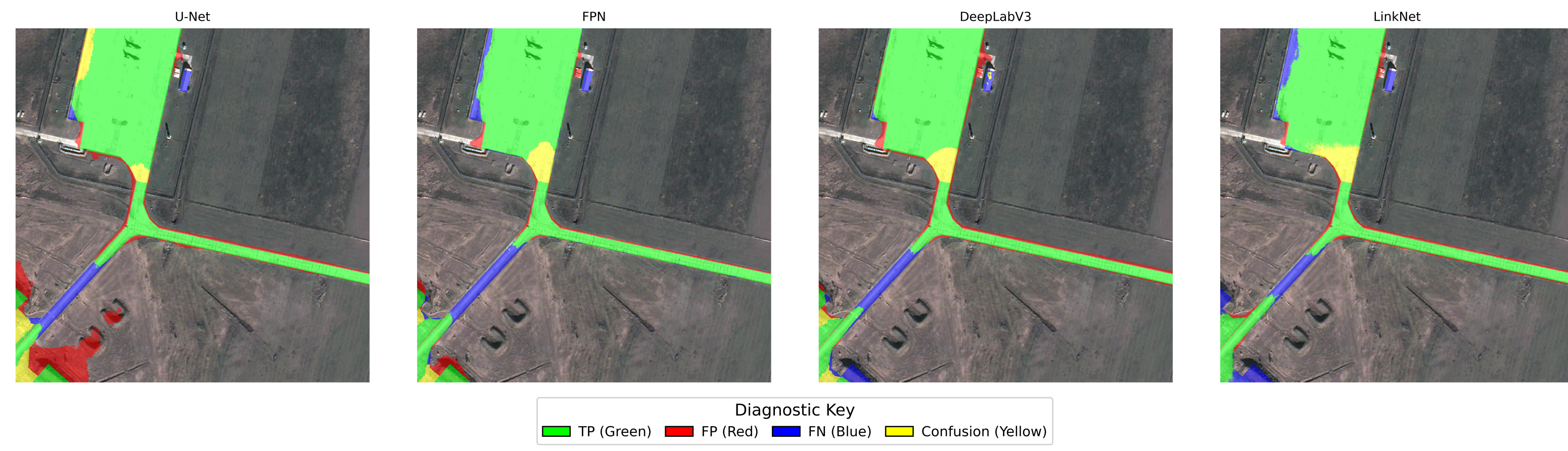
Iter.	IoU	Dice	PPV	AUC	Kappa	MCC
1	0.3483	0.4886	0.6075	0.7435	0.4857	0.5885
2	0.3661	0.4977	0.6630	0.7372	0.4952	0.6042
3	0.3922	0.5246	0.6943	0.7429	0.5229	0.6322
4	0.3928	0.5248	0.7080	0.7423	0.5225	0.6350



RQ2. Which segmentation model performs better?

Model	IoU	Dice	PPV	AUC	Kappa	MCC
U-Net	0.3928	0.5248	0.7080	0.7423	0.5225	0.6350
FPN	0.3934	0.5243	0.7222	0.7417	0.5222	0.6379
DeepLabV3	0.3895	0.5248	0.7089	0.7424	0.5224	0.6353
LinkNet	0.2024	0.2910	0.3130	0.6748	0.2862	0.3014

4-Model Comparison (1x4, Classic Colors) for AOI_063_20211122_HR | Patch R:5, C:1, Cell:1000px



Conclusions & Future Work

- Geo-SEAM turns noisy, open-source vector annotations into **high-quality training masks**.
- Active learning achieved a **12x speedup** for the **dataset creation**.
- A **user-friendly UI** and **highly scalable framework** make the human-in-the-loop process fast and accessible.

Future Work

- Use the complete **155 image dataset** and compare against models trained on fully hand-drawn masks.
- Use the generated masks to **fine-tune state-of-the-art segmentation architectures** like SAM, or Prithvi-EO.
- Migrate** the pipeline into **professional labeling software** for a better integration into the ML ecosystem

Selected References

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- [4] OpenStreetMap contributors, 2026. OpenStreetMap. [Online]. Available: <https://www.openstreetmap.org>.
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